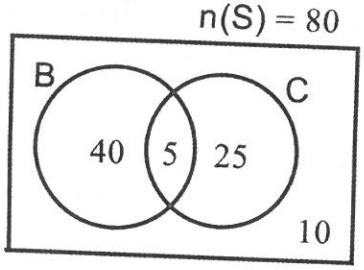
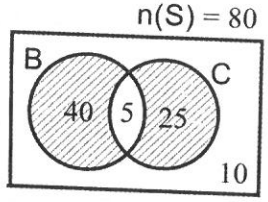
1.2.2



(5)

1.2.3 $P(\text{only one}) = \frac{40+25}{80} = \frac{65}{80} = \frac{13}{16}$



(3)

[22]

QUESTION 2

2.1 $P(\text{not (A or B)}) = 1 - 0{,}9 = 0{,}1$

(2)

2.2 $P(\text{A or B}) = P(\text{A}) + P(\text{B}) - P(\text{A and B})$

$\therefore 0{,}9 = 0{,}38 + 0{,}52 - P(\text{A and B})$

$\therefore P(\text{A and B}) = 0{,}38 + 0{,}52 - 0{,}9$

$= 0$

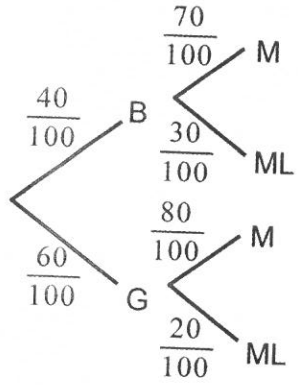
(4)

2.3.1 Yes, A and B are mutually exclusive because $P(\text{A and B}) = 0$. (2)

2.3.2 No, A and B are not complementary because $P(\text{A or B}) \neq 1$. (2)

[10]

3.1



(5)

3.2 P(learner who takes ML)

$= P((\text{B and ML}) \text{ or } (\text{G and ML}))$

$= \left(\frac{40}{100} \times \frac{30}{100}\right) + \left(\frac{60}{100} \times \frac{20}{100}\right)$

$= \frac{6}{25}$

(5)

[10]

QUESTION 4

4.1 $P(\text{girl's name first}) = \frac{2}{5}$

(1)

4.2 $P(\text{boy's name second}) = \frac{3}{4}$

(2)

4.3 $P(\text{GG or BB}) = \left(\frac{2}{5} \times \frac{1}{4}\right) + \left(\frac{3}{5} \times \frac{2}{4}\right)$

$= \frac{2}{20} + \frac{6}{20} = \frac{8}{20} = \frac{2}{5}$

(5)

[8]

TOTAL: 50